

Instructional improvement data

West, Welch, and Galecki (2007) analyzed a dataset on first- and third-grade students from the Study of Instructional Improvement (SII) by Hill, Rowan and Ball (2005).

The question of interest is how teachers' experience, mathematics preparation, and mathematics content knowledge affect students' gain in mathematics achievement scores from kindergarten to first grade.

The variables in the dataset `instruction.dta` used here are

- Level 1 (students)
 - `childid`: student identifier
 - `mathgain`: gain in math achievement score from spring of kindergarten to spring of first grade
 - `mathkind`: math achievement score in the spring of the kindergarten year
 - `girl`: dummy variable for being a girl
 - `minority`: dummy variable for being a minority student
 - `ses`: socioeconomic status
- Level 2 (class)
 - `classid`: class identifier
 - `yearstea`: first-grade teacher's years of teaching experience
 - `mathprep`: first-grade teacher's mathematics preparation (score based on number of mathematics content and methods courses)
 - `mathknow`: first-grade teacher's mathematics content knowledge, based on a 30-item scale with higher values indicating better knowledge
- Level 3 (school)
 - `schoolid`: school identifier
 - `housepov`: percentage of households in the neighborhood of the school below the poverty level